Topic 2.13 - Exponential and Logarithmic Equations and Inequalities

Guided Notes

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: ____________ Class: ____________ Date: ____________

Why this topic matters

Exponential and logarithmic equations are solved by exposing the inverse structure. Exact solutions often use common bases or logarithms; approximate solutions use technology only after a correct algebraic setup. Logarithmic equations require a final domain check because algebraic manipulation can create invalid candidates.

Learning targets

- Solve exponential equations and inequalities exactly or approximately.
- Solve logarithmic equations and inequalities while enforcing domain restrictions.
- Interpret solutions and thresholds in transformed functions and contextual models.

Language and structure

Term	Meaning in this topic
common-base method	Rewrite both sides with the same exponential base and equate exponents.
logarithmic method	Apply a logarithm to isolate an exponent.
extraneous solution	A candidate produced algebraically that violates an original equation or domain.
monotonic inequality	An inequality solved using whether an exponential or logarithmic function is increasing or decreasing.
threshold	An input at which a modeled quantity first reaches or passes a specified level.

Launch: make a claim before calculating

Launch investigation

Reasoning first

Compare the strategies for solving $2^{x+1} = 7$ and $2^{x+1} = 16$. Which solution can be exact without logarithms, and why?

Concept 1: Exponential equations

Core idea

Use common bases when possible; otherwise isolate the exponential and apply a logarithm.

- If $b^u = b^v$ with $b > 0$, $b \neq 1$, then $u = v$.
- $b^x = c$ gives $x = \log_b c = \ln c / \ln b$.
- Substitution such as $u = b^x$ can turn some equations into quadratics.

Worked example - annotate the reasoning

Problem. Solve $3^{2x} - 10(3^x) + 9 = 0$.
Model reasoning. Let $u = 3^x > 0$. Then $u^2 - 10u + 9 = (u-1)(u-9) = 0$, so $u = 1$ or $u = 9$. Thus $x = 0$ or $x = 2$.

Check your understanding

Independent

Solve $5^{x-1} = 17$ exactly and approximately.

Concept 2: Logarithmic equations and domain checks

Core idea

Condense logs when helpful, convert to exponential form, solve, and test every candidate in the original domain.

- Each logarithm argument must be positive.
- Squaring or multiplying expressions can introduce candidates that do not satisfy the original equation.
- Check exact candidates before rounding.

Worked example - annotate the reasoning

Problem. Solve $\log_2(x-1) + \log_2(x-5) = 3$.
Model reasoning. Domain $x > 5$. Condense: $\log_2((x-1)(x-5)) = 3$, so $(x-1)(x-5) = 8$. This gives $x = 3 \pm 2\sqrt{3}$. Only $x = 3 + 2\sqrt{3}$ satisfies $x > 5$.

Check your understanding

Independent

Solve $\ln(x+2) - \ln(x-1) = \ln 3$.

Concept 3: Inequalities and thresholds

Core idea

Use monotonicity: bases greater than 1 preserve inequality direction; bases between 0 and 1 reverse it.

- Logarithms with base greater than 1 are increasing; bases between 0 and 1 are decreasing.
- Intersect the algebraic solution with the original domain.
- In discrete contexts, convert a real threshold time to the first allowed whole-number input.

Worked example - annotate the reasoning

Problem. Solve $(1/2)^{x-2} \geq 8$.
Model reasoning. Write $8 = (1/2)^{-3}$. Since base $1/2$ is decreasing, $x - 2 \leq -3$, so $x \leq -1$.

Check your understanding

Independent

A balance follows $B(t) = 2500(1.07)^t$. Find the first whole year when $B(t) > 4000$.

Synthesis and exit ticket

Before you leave

Use precise notation, show the invariant or inverse structure, and interpret each parameter in context. A correct number without a defensible method is incomplete.

Exit ticket 1

No calculator

Solve $4^{x-1} = 64$.

Exit ticket 2

No calculator

Solve $\log_3(x+1) = 2$.

Exit ticket 3

No calculator

Solve $\log_2(x-4) > 1$.